

# Production-Grade AI Fintech Platform Architecture for Indian Middle-Class Families

This document defines a production-grade AI-powered fintech platform focused on Indian middle-class families. The platform helps users with stock market investing, MCX trading insights, pension planning, NPS management, currency exchange tracking, budgeting, and AI-based financial education using low-cost cloud-native architecture.

## 1. Product Vision

The platform acts as an AI financial co-pilot that simplifies investing and financial planning for beginners and middle-class families.

Key goals:

- Goal-based investing
- AI-powered recommendations
- Secure end-to-end encrypted communication
- Budget optimization
- Retirement planning
- Real-time portfolio analytics
- Low infrastructure cost

## 2. Core Features

- AI Investment Assistant
- Goal-Based Investment Planning
- MCX Commodity Insights
- Stock Market Portfolio Tracking
- NPS & Pension Planning
- Forex Currency Exchange Monitoring
- Expense Tracking & Budget Analysis
- AI Financial Education in Simple Language
- Tax Saving Recommendations
- SIP Planning & Wealth Growth Forecasting
- Realtime Notifications using SignalR
- AI-based Risk Analysis

## 3. Target Users

- Salaried Employees
- Small Business Owners
- First-Time Investors
- Families Planning Retirement
- SIP Investors

- NPS Subscribers
- Commodity Traders

## 4. Technology Stack

Frontend:

- Angular 19
- Tailwind CSS
- Chart.js

Backend:

- ASP.NET Core 9
- Clean Architecture
- CQRS Pattern
- MediatR

Database:

- PostgreSQL
- Redis Cache

AI Layer:

- OpenAI APIs
- Semantic Kernel
- RAG Architecture
- Vector Database (Qdrant)

Realtime:

- SignalR

Messaging:

- RabbitMQ

Cloud:

- Azure App Services
- Azure Container Apps

Deployment:

- Docker

Monitoring:

- OpenTelemetry
- Grafana

Security:

- JWT Authentication
- AES-256 Encryption
- Signal Protocol for E2EE
- TLS 1.3
- RBAC

## 5. Production-Grade Architecture

Architecture Layers:

1. Frontend Layer
2. API Gateway Layer
3. Microservices Layer
4. AI Intelligence Layer
5. Data Layer
6. Realtime Communication Layer
7. Monitoring & Security Layer

Microservices:

- User Service
- Portfolio Service
- AI Recommendation Service
- Goal Planning Service
- Notification Service
- Market Data Service
- Pension/NPS Service
- Forex Service

## 6. End-to-End Encryption Design

The platform uses production-grade end-to-end encryption.

Key security features:

- Signal Protocol-based message encryption
- Device-based key generation
- AES-256 encrypted payloads
- TLS 1.3 transport security
- Private keys never leave user device
- Secure JWT token rotation
- API rate limiting
- Secure encrypted database storage

## 7. Low-Cost Cloud Architecture

Cost Optimization Strategies:

- Azure Container Apps instead of full Kubernetes initially
- PostgreSQL single server
- Redis basic tier
- Background jobs via RabbitMQ
- AI response caching
- Embedding caching
- Auto-scaling APIs
- CDN for frontend hosting
- Serverless scheduling jobs

Expected Initial Monthly Cost:

- Azure Hosting: ■4000–■8000
- OpenAI Usage: ■3000–■12000
- Database & Redis: ■3000

- Monitoring: ■1000

## 8. AI Features

AI Capabilities:

- Portfolio analysis
- Risk scoring
- Personalized SIP suggestions
- Retirement planning
- Budget optimization
- Tax-saving guidance
- AI-generated investment explanations
- Market news summarization
- AI financial chatbot

RAG Data Sources:

- RBI Data
- NSE/BSE Data
- Government Pension Schemes
- SEBI Rules
- Financial Articles

## 9. MCX & Commodity Trading Features

- Commodity Watchlists
- Gold/Silver/Copper Analysis
- Price Alerts
- AI Commodity Trend Predictions
- Volatility Analysis
- Risk Warnings
- Historical Price Analytics

## 10. Stock Market Features

- Portfolio Dashboard
- Real-Time Profit/Loss Tracking
- Sector Exposure Analysis
- AI Risk Detection
- SIP Recommendations
- Long-Term Wealth Planning
- AI Stock Explanation Engine

## 11. NPS & Pension Features

- Retirement Goal Calculator
- NPS Fund Analysis
- Tax Benefit Suggestions
- Pension Corpus Forecasting
- Monthly Retirement Income Estimation
- Government Scheme Comparison

## **12. Forex & Currency Exchange**

- Currency Rate Tracking
- USD/INR Alerts
- Forex Analytics
- AI Currency Trend Insights
- International Travel Expense Planning
- Remittance Optimization

## **13. Database Design**

Main Tables:

- Users
- Goals
- Investments
- Transactions
- AIConversations
- PensionPlans
- CommodityTrades
- CurrencyRates
- Notifications

## **14. DevOps & CI/CD**

CI/CD Pipeline:

- GitHub Actions
- Docker Build
- Automated Testing
- Security Scanning
- Deployment Automation
- Rollback Strategy
- Environment-based Configuration

## **15. Monitoring & Observability**

Monitoring Stack:

- OpenTelemetry
- Grafana
- Azure Application Insights

Metrics:

- API Latency
- AI Token Usage
- User Activity
- Queue Performance
- Database Performance
- Failed Requests

## **16. Scalability Design**

Scalability Features:

- Horizontal API scaling
- Redis caching
- Queue-based async processing
- Read replicas for PostgreSQL
- AI response caching
- Background workers
- CDN-based frontend delivery

## **17. Resume & Startup Value**

This project demonstrates:

- AI Engineering
- .NET Enterprise Architecture
- Distributed Systems
- Fintech Domain Knowledge
- Secure End-to-End Encryption
- Cloud-Native Deployment
- Event-Driven Systems
- Realtime Communication

Strong positioning for:

- Product Companies
- GCCs
- Fintech Startups
- AI Startups
- Remote International Roles

## **18. Recommended MVP**

Phase 1:

- Authentication

- Portfolio Dashboard
- AI Chatbot
- Goal Planner
- Budget Tracker

Phase 2:

- MCX Integration
- NPS Planner
- Forex Analytics
- AI Recommendations

Phase 3:

- RAG Architecture
- Advanced Analytics
- Realtime Alerts
- AI Agents

## Final Conclusion

Final Recommendation:

This platform should focus on:

- Simplicity
- Financial literacy
- AI-powered personalization
- Security
- Low infrastructure cost
- Production-grade architecture

This combination of AI + Fintech + .NET + Cloud + End-to-End Encryption creates a highly scalable and market-relevant platform.